2026 PROJECT 2 PRG1781 [70 MARKS]

Problem Statement: Library Book Management System

You are tasked with developing a basic **Library Book Management System** using core programming principles. The program should allow users to:

1. Add books to the library.
2. Search for a book by title or author.
3. Borrow a book.
4. Return a book.
5. Display all available books.

Requirements

a) The system should be **menu-driven** and run continuously until the user decides to exit.

b) The program should handle invalid inputs gracefully.

c) Books should be represented as objects or as structured records (depending on the language used).

d) Maintain a collection of books, where each book has:

   1. **Title**
   2. **Author**
   3. **Unique ID**
   4. **Status**: "Available" or "Borrowed"

e) When a user borrows a book, the program should:

   1. Change the status of the book to "Borrowed."
   2. Prevent the user from borrowing it again until it is returned.

f) When a user returns a book, the status should update back to "Available."

PROJECT [QUESTION 1]

Based on the description above, answer the questions below:

1. Explain how you will design the logic for the Library Book Management System. Your explanation should cover:
   - How you will store book records (Array, List, Dictionary, etc.).
   - How books will be added, searched, borrowed, and returned.
   - How you will handle user input and errors.
   - How you will ensure books are not borrowed multiple times.

2. Write a structured pseudocode algorithm that outlines the step-by-step operations of the Library Book Management System. Your algorithm should:
   - Clearly define how the system starts.
   - Show how each menu function is handled.
   - Include validation checks for borrowing and returning books.

PROJECT [QUESTION2]

1. Draw a functional flowchart that represents the menu-driven structure of the Library Book Management System. Your flowchart should:

    - Start with "Display Menu".
    - Branch into the respective functions based on user input:
    - Add a book.
    - Search for a book.
    - Borrow a book.
    - Return a book.
    - Display all books.
    - Exit.
    - If the user chooses to exit, the program should terminate.

2. Ensure that your flowchart is well-structured, easy to follow, and adheres to standard flowchart conventions (start/end, processes, decisions, etc.).

IMPORTANT INFORMATION FOR PROJECT 2

- This is a group assignment. You will complete this assignment in groups of between 3 and 4 learners. **If you work on your own your mark will be capped at 50%**
- There will only be a single submission for each group. Further communication in this regard will be sent out during the week.
- Belgium Campus makes use of software that can **scan for plagiarism** and a student caught doing this will get 0 for this assignment.
- Late assignments will not be accepted and missing the deadline is an automatic 0.

Criteria	Mark
1. Problem-Solving Logic & Explanation (How the system stores and processes book records, handling input validation, error handling)	15
2. Overall Structure of the Pseudocode Algorithm (Completeness, clarity, logic flow)	15
3. Functional Flowchart (Logical representation of the menu-driven system, correctness)	15
4. Flowchart Presentation & Clarity (Proper use of flowchart symbols, readability)	10
5. Implementation Feasibility (How well the logic translates into an actual programming solution)	5

6. Overall Project Presentation (Team explanation, clarity of concepts, answering questions)	10
Total:	70